Supplement 4: Theophylline Model

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Introduction

We have assumed the same PK and statistical models to model the theophylline data as were used in [1] so that the only differences are in the horseshoe part of the model specification.

Note: We have done the same for the remifentanil, tobramycin, and warfarin models.

PK Model

We assume a one-compartment model with first-order absorption and elimination with parameters (CL, V_c, K_a) representing clearance, central compartment volume, and absorption rate constant. The corresponding system of ODEs is:

$$\begin{aligned} \frac{dA_d}{dt} &= -K_a A_d \\ \frac{dA_c}{dt} &= K_a A_d - \left(\frac{CL}{V_c}\right) A_c \end{aligned} \tag{1}$$

where A_d is the amount in the depot and A_c is the amount in the central compartment. After solving this system of ODEs, the concentration is then $\frac{A_c}{V_c}$.

Statistical Model

We assume a model with proportional-plus-additive residual error:

$$C_{ij} = f\left(\boldsymbol{\theta}_i,\ t_{ij}\right) \times \left(1 + \epsilon_{p_{ij}}\right) + \epsilon_{a_{ij}}$$

where C_{ij} is the observed concentration at time t_{ij}, $\boldsymbol{\theta}_i = [CL_i, V_{c_i}, K_{a_i}]^\top$ is a vector containing the individual clearance, central compartment volume, and absorption rate constant for individual i, respectively, $f(\cdot,\cdot)$ is the one-compartment model mean function that is the solution to the system of ODEs in Equation 1, and ϵ_p and ϵ_a are normally distributed errors.

We directly model the *standard deviations* (ω_{CL}, ω_{V_c}, ω_{K_a}) of the random effects and assume a diagonal covariance matrix Ω such that

$$
\begin{aligned}
\Omega &= \begin{pmatrix} \omega_{CL}^2 & 0 & 0 \\ 0 & \omega_{V_c}^2 & 0 \\ 0 & 0 & \omega_{K_a}^2 \end{pmatrix} \\
&= \begin{pmatrix} \omega_{CL} & 0 & 0 \\ 0 & \omega_{V_c} & 0 \\ 0 & 0 & \omega_{K_a} \end{pmatrix} \begin{pmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{pmatrix} \begin{pmatrix} \omega_{CL} & 0 & 0 \\ 0 & \omega_{V_c} & 0 \\ 0 & 0 & \omega_{K_a} \end{pmatrix}
\end{aligned}
$$

We have used relatively weakly informative priors for all parameters. The full statistical model is:

$$
\begin{aligned}
C_{ij} \mid \boldsymbol{\theta},\ \boldsymbol{\eta}_i,\ \boldsymbol{\Omega},\ \sigma_p,\ \sigma_a &\sim Normal\left(f(\boldsymbol{\theta}_i,\ t_{ij}),\ \sigma_{ij}\right) \\
\boldsymbol{\eta}_i \mid \boldsymbol{\Omega} &\sim Normal\left(\begin{pmatrix} 0 \\ 0 \\ 0 \end{pmatrix},\ \boldsymbol{\Omega}\right) \\
TVCL &\sim Lognormal\left(log\left(1\right),\ 1\right) \\
TVVC &\sim Lognormal\left(log\left(8\right),\ 1\right) \\
TVKA &\sim Lognormal\left(log\left(0.8\right),\ 1\right) I\left(TVKA > \frac{TVCL}{TVVC}\right) \\
\omega_{CL} &\sim Half - Normal(0,\ 0.4) \\
\omega_{V_c} &\sim Half - Normal(0,\ 0.4) \\
\omega_{K_a} &\sim Half - Normal(0,\ 0.4) \\
\sigma_p &\sim Half - Normal(0,\ 0.5) \\
\sigma_a &\sim Half - Normal(0,\ 0.5) \\
\theta_{P_j} \mid \tau,\ \lambda_{P_j},\ c &\sim Normal(0,\ \tau\tilde{\lambda}_{P_j}),\ P \in \{CL, V_c, K_a\}, j = 1, \ldots, d_{cov} \\
\lambda_{P_j} &\sim Half - Cauchy(0, 1),\ P \in \{CL, V_c, K_a\}, j = 1, \ldots, d_{cov} \\
\tau &\sim Half - Cauchy(0, 0.032) \\
c^2 &\sim Inv - Gamma({}^{3}/_{2}, {}^{(3)(0.5^2)}/_{2}) \\
\tilde{\lambda}_{P_j}^2 &:= \frac{c^2 \lambda_{P_j}^2}{c^2 + \tau^2 \lambda_{P_j}^2}
\end{aligned}
$$

where

$$
\begin{aligned}
\boldsymbol{\theta} &= \begin{pmatrix} TVCL \\ TVVC \\ TVKA \end{pmatrix} \\
\boldsymbol{\eta}_i &= \begin{pmatrix} \eta_{CL_i} \\ \eta_{V_{c_i}} \\ \eta_{K_{a_i}} \end{pmatrix} \\
\boldsymbol{\theta}_i &= \begin{pmatrix} CL_i \\ V_{c_i} \\ K_{a_i} \end{pmatrix} = \begin{pmatrix} TVCL \times e^{\boldsymbol{X}_i \boldsymbol{\theta}_{CL} + \eta_{CL_i}} \\ TVVC \times e^{\boldsymbol{X}_i \boldsymbol{\theta}_{V_c} + \eta_{V_{c_i}}} \\ TVKA \times e^{\boldsymbol{X}_i \boldsymbol{\theta}_{K_a} + \eta_{K_{a_i}}} \end{pmatrix} \\
\sigma_{ij} &= \sqrt{f(\boldsymbol{\theta}_i,\ t_{ij})^2 \times \sigma_p^2 + \sigma_a^2}
\end{aligned}
$$

$\boldsymbol{X}_i$ is the ith row of $\boldsymbol{X}$, the $n_{subjects} \times d_{cov}$ design matrix, and d_{cov} is the number of covariates ($d_{cov} = 2$ in this dataset).

Note: The indicator for $TVKA$ is a constraint on the population value of absorption rate to avoid identifiability issues from flip-flop kinetics.

Note: The above univariate normal distributions (including lognormal and half-normal) are parameterized with the *standard deviation* rather than the variance to remain consistent with the parameterization that Stan (and R) use. The multivariate normal is parameterized with the mean and covariance matrix.

Note: d_{cov} can differ between parameters in a case where a covariate is obviously only relevant for a subset of the parameters (*e.g.* fed/fasted is relevant for absorption, but not for clearance), and $\boldsymbol{X}$ can be modified accordingly.

Data

The data can be seen below:

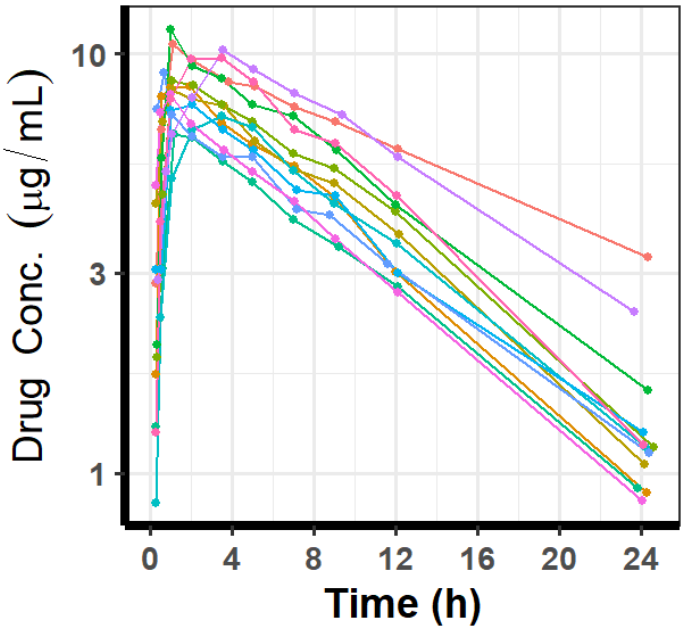

Figure S15: Theophylline data

References

[1] Géraldine Ayral, Jean-François Si Abdallah, Claude Magnard, and Jonathan Chauvin. A novel method based on unbiased correlations tests for covariate selection in nonlinear mixed effects models: The cossac approach. *CPT: pharmacometrics & systems pharmacology*, 10(4):318–329, 2021.